

HIV Dashboard Indicator Selection Report

This report documents which candidate indicators are included or excluded from the HIV correlation heatmaps in the dashboard.

Inclusion rule: indicators are shown when their average absolute Pearson correlation across HIVI_2019 through HIVI_2023 is ≥ 0.15 .

Interpretation: this is exploratory correlation, not causal proof. Indicators are used to prioritize patterns for public-health action, follow-up analysis, and policy discussion.

Count-based crime and legal-system measures are retained only in the Community Stress heatmap and should be read as community stress, enforcement, reporting, and structural-risk proxies rather than direct causes.

Important exclusions and cautions:

? Weak correlations are excluded from the heatmap because they can distract from stronger patterns and invite overinterpretation.

? CDP_2017-2021 is not used as a poverty/deprivation variable because the source dictionary identifies it as prostate cancer diagnosis count.

? Race variables are structural context variables. They should be framed as evidence of segregation and disinvestment patterns, not as biological or individual-level risk claims.

Threshold: average $|r| \geq 0.15$ | Page 1

Selection Summary

Priority Factors for City Action

Included: 13 indicators. Excluded: 17 indicators.

Highest included signals:

- ? Owner occupied housing (%) (Housing & neighborhood): avg |r| 0.440, negative.
- ? Drug abuse crimes (Behavioral & community risk): avg |r| 0.301, positive.
- ? PM2.5 concentration (Housing & neighborhood): avg |r| 0.262, negative.
- ? Social Engagement Index (Structural inequity & race): avg |r| 0.253, negative.
- ? Primary care provider access (%) (Healthcare access): avg |r| 0.188, negative.
- ? Opioid overdose mortality rate (Behavioral & community risk): avg |r| 0.186, positive.

Community Stress & Structural Risk

Included: 16 indicators. Excluded: 8 indicators.

Highest included signals:

- ? Crime - Battery (Community stress & safety): avg |r| 0.649, positive.
- ? Juvenile / youth behavioral indicator (Community stress & safety): avg |r| 0.644, positive.
- ? Youth juvenile court / community outcomes (Community stress & safety): avg |r| 0.626, positive.
- ? Crime - Murder / Manslaughter (Community stress & safety): avg |r| 0.625, positive.
- ? Incarceration / drug use / legal involvement (Structural poverty & opportunity): avg |r| 0.610, positive.
- ? Child poverty / community resource indicator (Structural poverty & opportunity): avg |r| 0.609, positive.

Full Selected Indicator Heatmap

Included: 5 indicators. Excluded: 11 indicators.

Highest included signals:

- ? PM2.5 concentration (Housing & neighborhood): avg |r| 0.262, negative.
- ? Primary care provider (%) (Healthcare access): avg |r| 0.188, negative.
- ? Rent-burdened households (%) (Housing & neighborhood): avg |r| 0.186, positive.
- ? Uninsured rate (%) (Healthcare access): avg |r| 0.177, positive.
- ? Median household income (Economic factors): avg |r| 0.154, negative.

Threshold: average |r| >= 0.15 | Page 2

Priority Factors for City Action

Included Indicators (13)

- ? Owner occupied housing (%) [Housing & neighborhood]: avg $|r|$ 0.440. Included because avg $|r|$ = 0.440, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Housing stability / ownership
- ? Drug abuse crimes [Behavioral & community risk]: avg $|r|$ 0.301. Included because avg $|r|$ = 0.301, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Drug-related enforcement proxy; count, interpret cautiously
- ? PM2.5 concentration [Housing & neighborhood]: avg $|r|$ 0.262. Included because avg $|r|$ = 0.262, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Environmental exposure
- ? Social Engagement Index [Structural inequity & race]: avg $|r|$ 0.253. Included because avg $|r|$ = 0.253, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Civic/social isolation and built environment
- ? Primary care provider access (%) [Healthcare access]: avg $|r|$ 0.188. Included because avg $|r|$ = 0.188, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Routine access to care
- ? Opioid overdose mortality rate [Behavioral & community risk]: avg $|r|$ 0.186. Included because avg $|r|$ = 0.186, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Opioid mortality
- ? Rent-burdened households (%) [Housing & neighborhood]: avg $|r|$ 0.186. Included because avg $|r|$ = 0.186, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Housing affordability stress
- ? Drug overdose mortality rate [Behavioral & community risk]: avg $|r|$ 0.181. Included because avg $|r|$ = 0.181, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Drug-related mortality
- ? Social Vulnerability Index [Housing & neighborhood]: avg $|r|$ 0.177. Included because avg $|r|$ = 0.177, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Neighborhood vulnerability
- ? Uninsured rate (%) [Healthcare access]: avg $|r|$ 0.177. Included because avg $|r|$ = 0.177, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Coverage barrier
- ? Food insecurity (%) [Economic factors]: avg $|r|$ 0.176. Included because avg $|r|$ = 0.176, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Material hardship and basic needs
- ? Black or African American residents (%) [Structural inequity & race]: avg $|r|$ 0.166. Included because avg $|r|$ = 0.166, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Race concentration proxy; reflects segregation and structural context, not individual causation
- ? Median household income [Economic factors]: avg $|r|$ 0.154. Included because avg $|r|$ = 0.154, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Income resources

Priority Factors for City Action (continued)

Excluded Indicators (17)

- ? Arts education access and quality (%) [Education & opportunity]: avg $|r|$ 0.150. Excluded from heatmap because avg $|r|$ = 0.150, below the 0.15 threshold; correlation is too weak or unstable for the visual. CPS arts education access/quality
- ? Poverty rate (%) [Economic factors]: avg $|r|$ 0.146. Excluded from heatmap because avg $|r|$ = 0.146, below the 0.15 threshold; correlation is too weak or unstable for the visual.
Material hardship
- ? Hispanic or Latino residents (%) [Structural inequity & race]: avg $|r|$ 0.146. Excluded from heatmap because avg $|r|$ = 0.146, below the 0.15 threshold; correlation is too weak or unstable for the visual. Ethnicity concentration proxy; structural context, not individual causation
- ? White non-Hispanic residents (%) [Structural inequity & race]: avg $|r|$ 0.144. Excluded from heatmap because avg $|r|$ = 0.144, below the 0.15 threshold; correlation is too weak or unstable for the visual. Race/segregation proxy; negative correlation can indicate concentration in non-white/disinvested areas
- ? Primary care physicians per capita [Healthcare access]: avg $|r|$ 0.143. Excluded from heatmap because avg $|r|$ = 0.143, below the 0.15 threshold; correlation is too weak or unstable for the visual. Provider supply
- ? Child Opportunity Index - education domain [Education & opportunity]: avg $|r|$ 0.136. Excluded from heatmap because avg $|r|$ = 0.136, below the 0.15 threshold; correlation is too weak or unstable for the visual. Neighborhood education opportunity for children
- ? Unemployment rate (%) [Economic factors]: avg $|r|$ 0.133. Excluded from heatmap because avg $|r|$ = 0.133, below the 0.15 threshold; correlation is too weak or unstable for the visual.
Job access and labor-market exclusion
- ? Severely rent-burdened households (%) [Housing & neighborhood]: avg $|r|$ 0.126. Excluded from heatmap because avg $|r|$ = 0.126, below the 0.15 threshold; correlation is too weak or unstable for the visual. Severe housing affordability stress
- ? Child Opportunity Index [Structural inequity & race]: avg $|r|$ 0.117. Excluded from heatmap because avg $|r|$ = 0.117, below the 0.15 threshold; correlation is too weak or unstable for the visual. Childhood opportunity / structural investment
- ? Area Deprivation Index [Economic factors]: avg $|r|$ 0.116. Excluded from heatmap because avg $|r|$ = 0.116, below the 0.15 threshold; correlation is too weak or unstable for the visual.
Neighborhood deprivation including income, education, employment, and housing quality
- ? Physical inactivity (%) [Behavioral & community risk]: avg $|r|$ 0.106. Excluded from heatmap because avg $|r|$ = 0.106, below the 0.15 threshold; correlation is too weak or unstable for the visual. Behavioral risk marker
- ? Adult smoking rate (%) [Behavioral & community risk]: avg $|r|$ 0.085. Excluded from heatmap because avg $|r|$ = 0.085, below the 0.15 threshold; correlation is too weak or unstable for

Priority Factors for City Action (continued)

the visual. Behavioral risk marker

? College graduation rate (%) [Education & opportunity]: avg $|r|$ 0.084. Excluded from heatmap because avg $|r|$ = 0.084, below the 0.15 threshold; correlation is too weak or unstable for the visual. Adult educational attainment

? Adult obesity rate (%) [Behavioral & community risk]: avg $|r|$ 0.082. Excluded from heatmap because avg $|r|$ = 0.082, below the 0.15 threshold; correlation is too weak or unstable for the visual. Behavioral risk marker

? High school graduation rate (%) [Education & opportunity]: avg $|r|$ 0.074. Excluded from heatmap because avg $|r|$ = 0.074, below the 0.15 threshold; correlation is too weak or unstable for the visual. Adult educational attainment

? Preventive care routine checkup (%) [Healthcare access]: avg $|r|$ 0.073. Excluded from heatmap because avg $|r|$ = 0.073, below the 0.15 threshold; correlation is too weak or unstable for the visual. Prevention engagement

? Preschool enrollment (%) [Education & opportunity]: avg $|r|$ 0.069. Excluded from heatmap because avg $|r|$ = 0.069, below the 0.15 threshold; correlation is too weak or unstable for the visual. Early education access

Threshold: average $|r|$ $\geq$ 0.15 | Page 5

Community Stress & Structural Risk

Included Indicators (16)

- ? Crime - Battery [Community stress & safety]: avg $|r|$ 0.649. Included because avg $|r|$ = 0.649, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Crime count; use as community safety/enforcement proxy, not direct cause
- ? Juvenile / youth behavioral indicator [Community stress & safety]: avg $|r|$ 0.644. Included because avg $|r|$ = 0.644, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Youth behavioral/community stress proxy
- ? Youth juvenile court / community outcomes [Community stress & safety]: avg $|r|$ 0.626. Included because avg $|r|$ = 0.626, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Youth justice/community stress proxy
- ? Crime - Murder / Manslaughter [Community stress & safety]: avg $|r|$ 0.625. Included because avg $|r|$ = 0.625, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Crime count; use as severe community violence proxy
- ? Incarceration / drug use / legal involvement [Structural poverty & opportunity]: avg $|r|$ 0.610. Included because avg $|r|$ = 0.610, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Composite structural/legal involvement proxy
- ? Child poverty / community resource indicator [Structural poverty & opportunity]: avg $|r|$ 0.609. Included because avg $|r|$ = 0.609, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Child poverty/resource proxy
- ? Environmental air quality / pollution [Health & environment]: avg $|r|$ 0.603. Included because avg $|r|$ = 0.603, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Environmental exposure proxy
- ? Youth poverty / vulnerability index [Structural poverty & opportunity]: avg $|r|$ 0.594. Included because avg $|r|$ = 0.594, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Youth poverty/vulnerability proxy
- ? Crime - Robbery [Community stress & safety]: avg $|r|$ 0.585. Included because avg $|r|$ = 0.585, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Crime count; use as community safety/enforcement proxy
- ? Social wellbeing / youth wellness [Health & environment]: avg $|r|$ 0.502. Included because avg $|r|$ = 0.502, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Youth wellness/social wellbeing proxy
- ? Racial / income poverty zone indicator [Structural poverty & opportunity]: avg $|r|$ 0.441. Included because avg $|r|$ = 0.441, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Structural inequity proxy
- ? ZIP code quality / clinical care index [Access & clinical context]: avg $|r|$ 0.424. Included because avg $|r|$ = 0.424, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Clinical care quality/access proxy
- ? Low-access / food access index [Structural poverty & opportunity]: avg $|r|$ 0.398. Included because avg $|r|$ = 0.398, meeting the ≥ 0.15 exploratory signal threshold. Direction is

Community Stress & Structural Risk (continued)

negative. Food access / neighborhood access proxy

? Alcohol / drug problem prevalence [Substance use & mental health]: avg $|r|$ 0.295. Included because avg $|r|$ = 0.295, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Rate measure preferred over count

? Primary care provider access prevalence [Access & clinical context]: avg $|r|$ 0.188. Included because avg $|r|$ = 0.188, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Rate measure preferred over count

? Health coverage / insurance prevalence [Access & clinical context]: avg $|r|$ 0.154. Included because avg $|r|$ = 0.154, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Rate measure preferred over count

Excluded Indicators (8)

? Unmet mental health need - stress/symptoms [Substance use & mental health]: avg $|r|$ 0.149. Excluded from heatmap because avg $|r|$ = 0.149, below the 0.15 threshold; correlation is too weak or unstable for the visual. Rate measure preferred over count

? Alcohol-induced mortality rate [Substance use & mental health]: avg $|r|$ 0.143. Excluded from heatmap because avg $|r|$ = 0.143, below the 0.15 threshold; correlation is too weak or unstable for the visual. Mortality rate

? No checkup / preventive care avoidance [Access & clinical context]: avg $|r|$ 0.126. Excluded from heatmap because avg $|r|$ = 0.126, below the 0.15 threshold; correlation is too weak or unstable for the visual. Rate measure preferred over count

? Recreational / illicit drug use prevalence [Substance use & mental health]: avg $|r|$ 0.095. Excluded from heatmap because avg $|r|$ = 0.095, below the 0.15 threshold; correlation is too weak or unstable for the visual. Rate measure preferred over count

? Obesity prevalence [Health & environment]: avg $|r|$ 0.082. Excluded from heatmap because avg $|r|$ = 0.082, below the 0.15 threshold; correlation is too weak or unstable for the visual. Rate measure preferred over count

? Routine checkup / preventive care prevalence [Access & clinical context]: avg $|r|$ 0.073. Excluded from heatmap because avg $|r|$ = 0.073, below the 0.15 threshold; correlation is too weak or unstable for the visual. Rate measure preferred over count

? Binge drinking / alcohol behavior prevalence [Substance use & mental health]: avg $|r|$ 0.070. Excluded from heatmap because avg $|r|$ = 0.070, below the 0.15 threshold; correlation is too weak or unstable for the visual. Rate measure preferred over count

? Unmet mental health need - adults [Substance use & mental health]: avg $|r|$ 0.067. Excluded from heatmap because avg $|r|$ = 0.067, below the 0.15 threshold; correlation is too weak or unstable for the visual. Rate measure preferred over count

Full Selected Indicator Heatmap

Included Indicators (5)

- ? PM2.5 concentration [Housing & neighborhood]: avg $|r|$ 0.262. Included because avg $|r|$ = 0.262, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Chicago Health Atlas Data Download - Community areas (8).xlsx
- ? Primary care provider (%) [Healthcare access]: avg $|r|$ 0.188. Included because avg $|r|$ = 0.188, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Chicago Health Atlas Data Download - Community areas (6).xlsx
- ? Rent-burdened households (%) [Housing & neighborhood]: avg $|r|$ 0.186. Included because avg $|r|$ = 0.186, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Chicago Health Atlas Data Download - Community areas (7).xlsx
- ? Uninsured rate (%) [Healthcare access]: avg $|r|$ 0.177. Included because avg $|r|$ = 0.177, meeting the ≥ 0.15 exploratory signal threshold. Direction is positive. Chicago Health Atlas Data Download - Community areas (6).xlsx
- ? Median household income [Economic factors]: avg $|r|$ 0.154. Included because avg $|r|$ = 0.154, meeting the ≥ 0.15 exploratory signal threshold. Direction is negative. Chicago Health Atlas Data Download - Community areas (12).xlsx

Excluded Indicators (11)

- ? Eviction rate (%) [Housing & neighborhood]: avg $|r|$ 0.147. Excluded from heatmap because avg $|r|$ = 0.147, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (7).xlsx
- ? Poverty rate (%) [Economic factors]: avg $|r|$ 0.146. Excluded from heatmap because avg $|r|$ = 0.146, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (12).xlsx
- ? Unemployment rate (%) [Economic factors]: avg $|r|$ 0.133. Excluded from heatmap because avg $|r|$ = 0.133, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (12).xlsx
- ? Environmental justice impact (%) [Housing & neighborhood]: avg $|r|$ 0.122. Excluded from heatmap because avg $|r|$ = 0.122, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (8).xlsx
- ? Physical inactivity (%) [Behavioral risk factors]: avg $|r|$ 0.106. Excluded from heatmap because avg $|r|$ = 0.106, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (12).xlsx
- ? Hardship index [Economic factors]: avg $|r|$ 0.105. Excluded from heatmap because avg $|r|$ = 0.105, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (12).xlsx
- ? Smoking prevalence (%) [Behavioral risk factors]: avg $|r|$ 0.085. Excluded from heatmap because avg $|r|$ = 0.085, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (13).xlsx

Full Selected Indicator Heatmap (continued)

? Obesity rate (%) [Behavioral risk factors]: avg $|r|$ 0.082. Excluded from heatmap because avg $|r|$ = 0.082, below the 0.15 threshold; correlation is too weak or unstable for the visual.

Chicago Health Atlas Data Download - Community areas (10).xlsx

? Overcrowded housing (%) [Housing & neighborhood]: avg $|r|$ 0.080. Excluded from heatmap because avg $|r|$ = 0.080, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (7).xlsx

? Preventive care: routine checkup (%) [Healthcare access]: avg $|r|$ 0.073. Excluded from heatmap because avg $|r|$ = 0.073, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (6).xlsx

? Per capita income [Economic factors]: avg $|r|$ 0.070. Excluded from heatmap because avg $|r|$ = 0.070, below the 0.15 threshold; correlation is too weak or unstable for the visual. Chicago Health Atlas Data Download - Community areas (12).xlsx

Threshold: average $|r|$ $\geq$ 0.15 | Page 9